ADVANTEST®

V93000 Specifications

V93000 SoC Pin Scale 9G Digital Card

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⚠ **Warning**	**Warning of a hazard or hazardous situation** A warning calls attention to a procedure, practice or similar process, which, if not correctly performed or adhered to, could result in death, injury or harm to personnel. ➡ Do not proceed beyond a warning unless the indicated conditions are fully understood and met.

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A caution calls attention to a procedure, practice or similar process, which, if not correctly performed or adhered to, could result in damage to or destruction of part or all of the test system equipment.

➡ Do not proceed beyond a caution unless the indicated conditions are fully understood and met.

PS9G specifications

The Pin Scale 9G (PS9G) is a high-speed digital card for the V93000 ATE platform.

Note	The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).

Contact information

For more information about the Advantest V93000 SoC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

PS9G overview

Scope of specifications

Advantest specifies and verifies the specifications at the DUT interface pogo level. Using an adequate DUT board, *valid system calibration* and a fixture delay measurement (TDR), these specifications are valid at the device under test as well. Specifications describe warranted product performance. Characteristics are included as typical values to provide additional useful information by describing typical non-warranted performance.

Test Processor Per-Pin architecture

Maximum channel count	64 channels per card
Parallel vector memory[1]	16 MByte (64 MVectors at x4 mode) standard, 32 MByte (128 MVectors at x4 mode) optional, 64 MByte (256 MVectors at x4 mode) optional, 112 MByte (448 MVectors at x4 mode) optional
Maximum scan depth[2]	28.5 GVectors (at x4mode)
PRBS per Pin	PRBS up to 2^32, programmable seed and compare User defined compliance and serial standard pattern space (32k per pin)
SmartLoop™	Active, Retimed, differential and single ended

Speed

Maximum data rate	> 8000 Mbps (x8 mode) 6800 Mbps (x6 mode)
Maximum clock rate	> 4 GHz

Maximum external input voltage (compliance range)

Maximum range for external voltage applied to pin[3]	-6.0V to +9.0V

Test programming and debug

The SmarTest production test environment includes manufacturing test interfaces to operators, handling equipment, and factory networks. A graphical testflow editor links device tests into a production-ready test program, where the tests are set up via fill-in-the-blank test methods. User-specific tests are programmed with test methods in C++. Links are available for design-to-test conversion. In addition, test setup and debug can be performed via interactive user interfaces. Result analysis tools are available for error locations, timing behavior, analog waveforms, shmoo plots, pin margin tests and memory bit map.

[1] *vector memory is used for pattern, setup data and result capture*
[2] *Maximum scan depth support total of 8 chains per instrument, 3.584GVector max (drive only, single bit mode) or 1.792 GVectors in x4 mode for EACH chain. Total data volume per instrument ~ 28.5 GVectors (in x4 mode).*
[3] *valid for DC connection; for AC connection -1V to 4V (50 Ohm termination)*

Compatibility

PS9G is supported starting with SmarTest 7.1 and higher. It can be used with existing Pin Scale, DC Scale, MBAV8, and Port Scale RF cards.

Calibration and operation

Warm-up time	60 minutes
Basic maintenance period	6 months
Base calibration period (traceable calibration)	6 months
Calibration period[4] (system auto adjustment)	3 months

Environmental

Operating	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	< 70% R. H., non-condensing
Storage (without water)	-40°C to +70°C

[4] *valid at ambient temperature within ±2.5 K of calibration temperature. For temperature requirements during calibration please see "Maintenance Guide."*

PS9G Timing

PS9G OTA/EPA

For OTA/EPA test method condition: TX to RX loop using PRBS 2^15 pattern and verified with Vswing = 250mV and 850mV and speeds up to 6.8 Gbps single ended mode and up to 8.0 Gbps differential mode, loop includes standard high speed FR4 DUT board traces

OTA/EPA Single Ended

	4.5 Gbps per pin pair	**4.5 Gbps board 64 pins**	**6.8 Gbps per pin pair**	**6.8 Gbps board 64 pins**
Overall Timing Accuracy (t_{OTA})	< ±35 ps	< ±46 ps	< ±44 ps	< ±55 ps
Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	< ±18 ps	< ±23 ps	< ±22 ps	< ±28 ps

OTA/EPA Differential

	4.5 Gbps per lane	**4.5 Gbps board 16 lanes**	**8 Gbps per lane**	**8 Gbps board 16 lanes**
Overall Timing Accuracy (t_{OTA})	< ±30 ps	< ±38 ps	< ±40 ps	< ±55 ps
Edge Placement Accuracy ($t_{EPA}=t_{OTA}/2$)	< ±15 ps	< ±19 ps	< ±20 ps	< ±28 ps

Edge Placement

Edge placement resolution	1 ps
Edge placement range[5]	-4 to 32 Sequencer periods

PS9G Vector Period

Minimum vector period[6]	125 ps
Vector Period Resolution	0.001 fs
Vector Period Accuracy	±15 ppm of period setting
Minimum sequencer period	880 ps
Maximum sequencer period	5000 ns

[5] *edge placement range maximal -5000 ns to +15000 ns*
[6] *x8 mode for differential drive and receive; over-programming possible*

PS9G Eye Height

Single Ended (verified with Vswing = 850mV into 50Ohm)

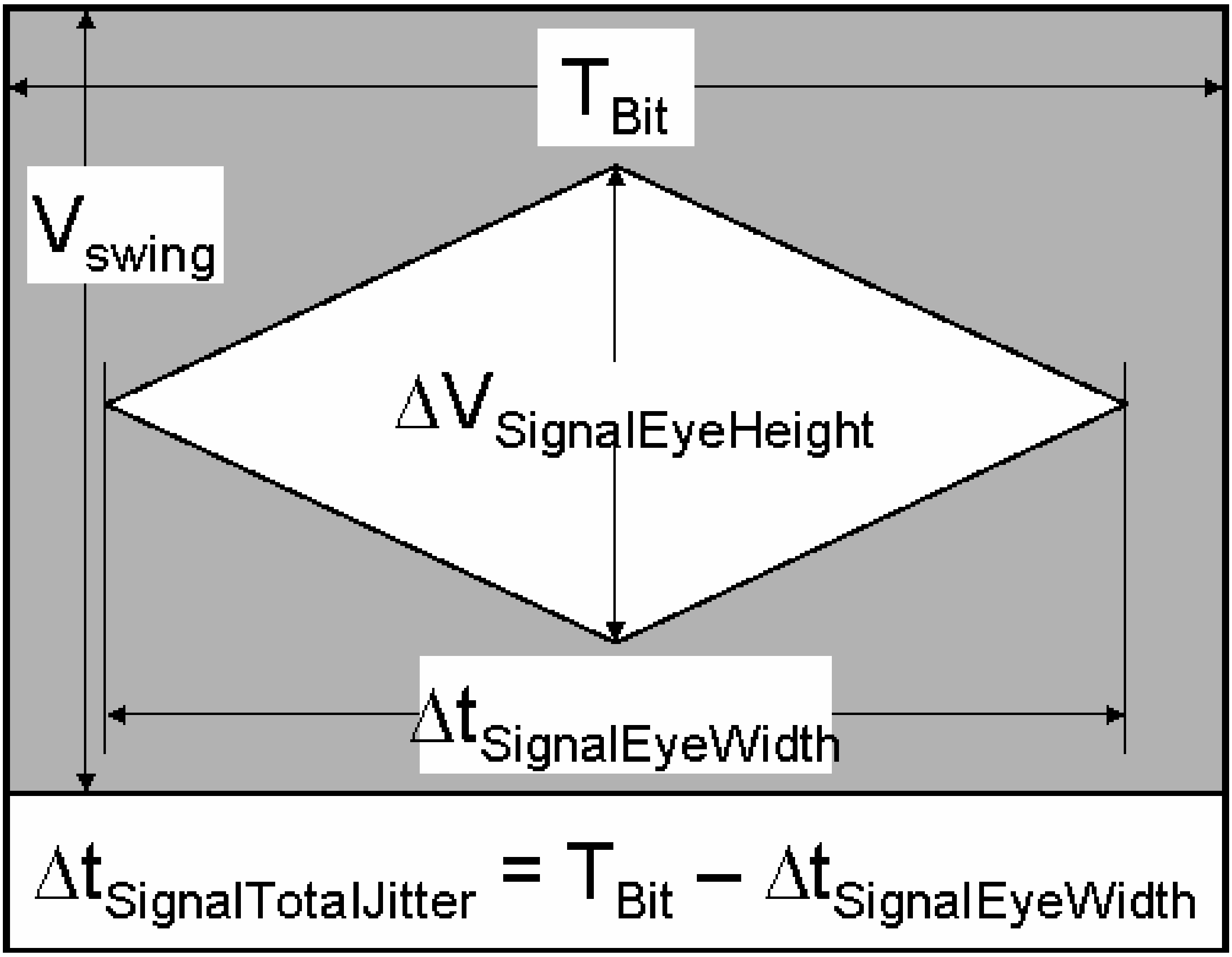
	2.5 Gbps	5.33 Gbps	6.8 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	89 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	88 % of V_{swing}	68 % of V_{swing}	54 % of V_{swing}
Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair[7]	78 % of V_{swing}	29 % of V_{swing}	15 % of V_{swing}

Differential (verified with Vswing = 250mV into 50Ohm)

	2.5 Gbps	5.33 Gbps	8.0 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	80 % of V_{swing}	68 % of V_{swing}	35 % of V_{swing}
Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair Minimum $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	74 % of V_{swing}	37 % of V_{swing}	11 % of V_{swing}

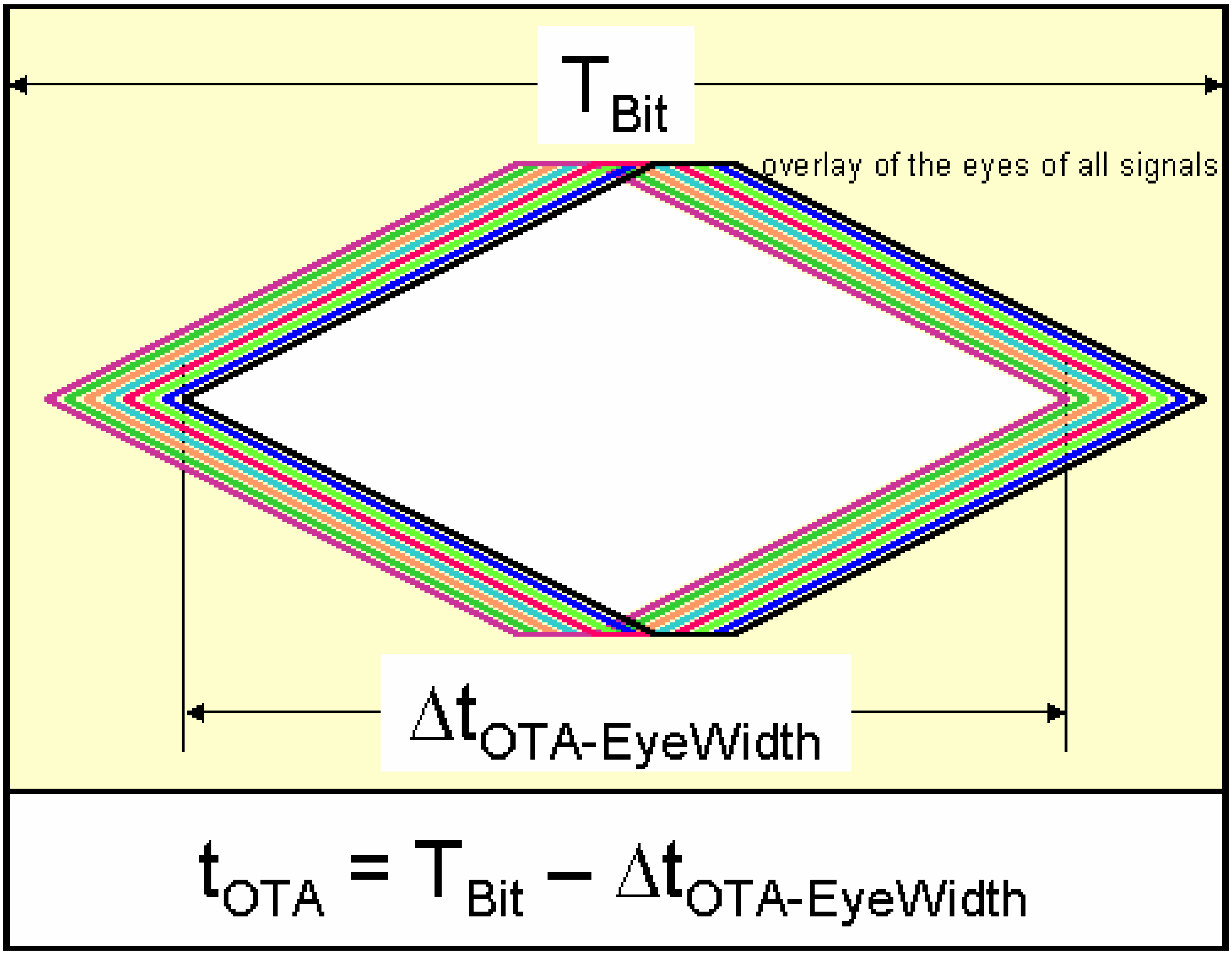
[7] *tester driver measured with tester receiver using PRBS 2^15 pattern, "equalization (Topic 91468)" is active.*

OTA Specification



Total Jitter (ΔtSignalTotalJitter)

Eye Diagram



Eye Diagram

PS9G Linearity

The graph below describes the non-linearities of the edge delays between the driver (programmed delay) and the receiver (measured delay) even after calibration.

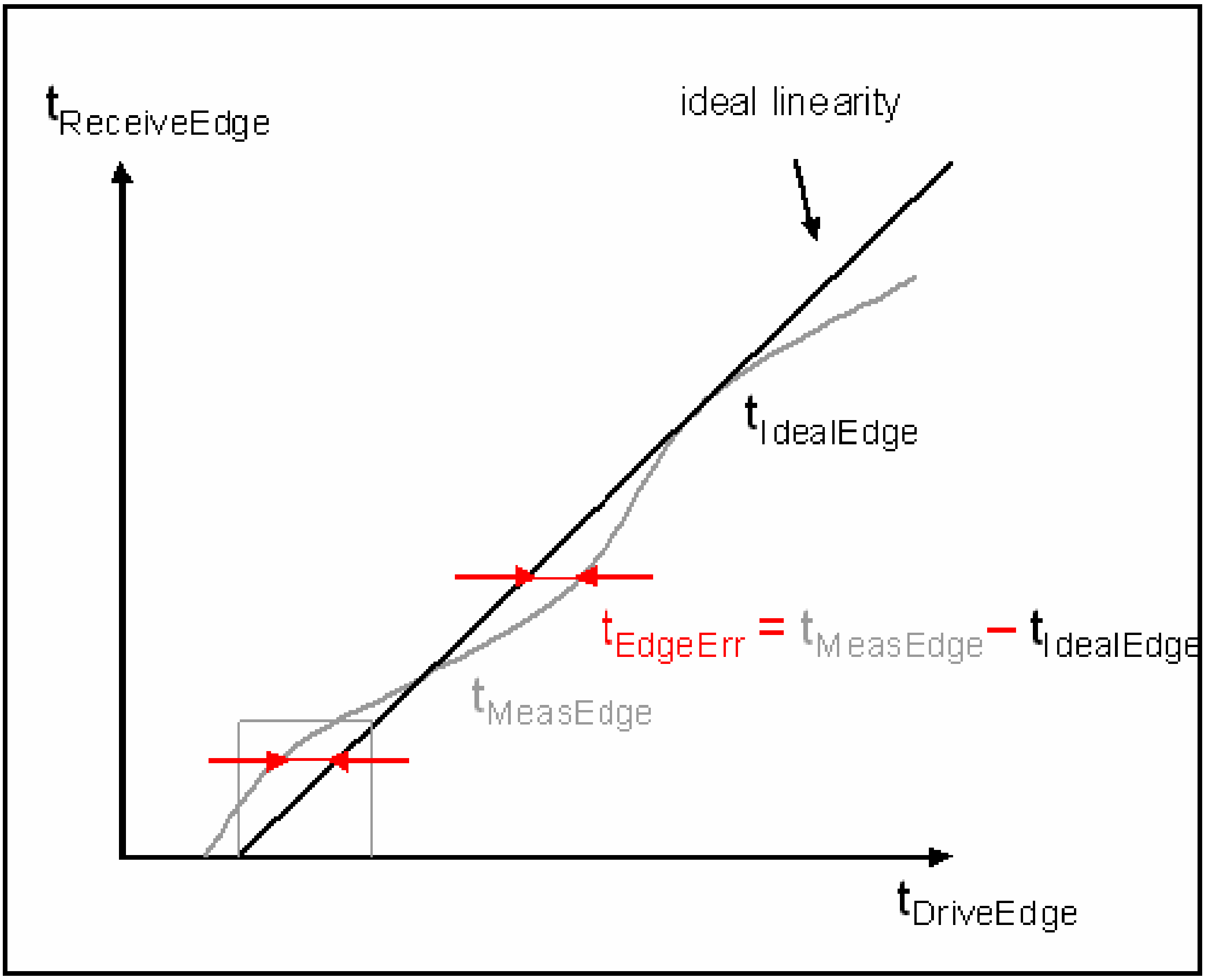
The black line represents the ideal delay, that is, the receiver delay follows the driver delay and may have an offset. The gray line represents a typical measurement. Here, the measured delay differs from the ideal delay by an edge delay offset ($t_{EdgeErr}$). This offset could be either a differential non-linearity (point error) or an integral non-linearity (integral error over a delay range).

The driver and receiver edges are treated independently and contribute equally to the uncertainty of the edge placement. Therefore, the edge placement non-linearity would be double of what is specified here.

Note These non-linearity terms are part of the *PS9G OTA/EPA* specifications and do not need to be considered separately.

Integral Non-Linearity[8] INL ($\lvert t_{EdgeErr} \rvert$)	±5 ps
Differential Non-Linearity DNL	5 ps

[8] *Single ended driver measured with single ended receiver and differential drive measured with differential receiver.*



PS9G Clocks

Differential clock[9] jitter at 4.0 GHz	typical 2.9 ps rms

PS9G Clock Data Recovery (CDR)

Tracking[10] range	±7.5 UI
Tracking max frequency without clock forwarding	100 MHz
Tracking max frequency with clock forwarding	1 MHz

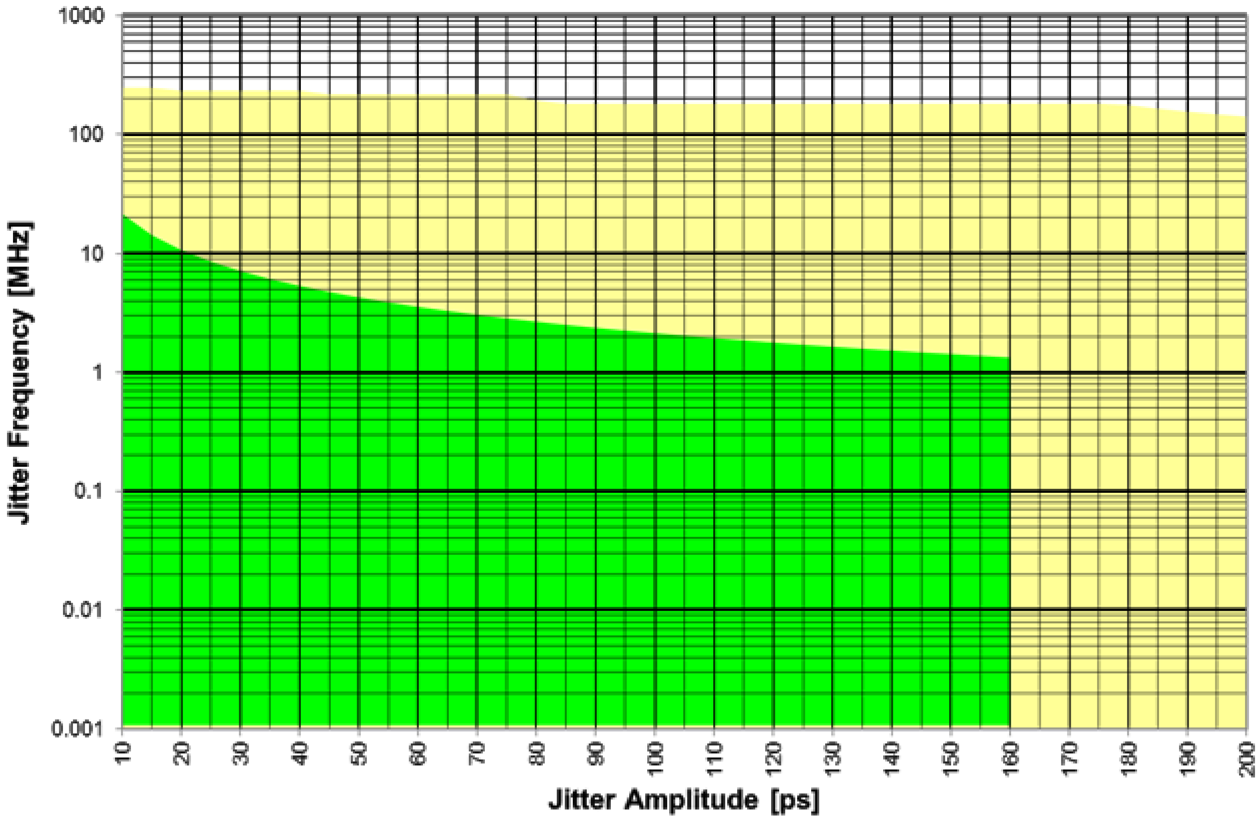
[9] *Tester driver measured with tester receiver and measured value divided by √2. Swing is in range 250mV..850mV*

[10] *Tracking operates in x4 and x6 mode, max speed limit is depending on high frequency jitter amplitude of device, this means typical speed limit is in range 3..4Gbps.*

PS9G Drive Jitter Injection

Jitter Injection via delay modulation with digital per pin AWG with 512 samples.

Timing Jitter range[11]	400 ps Peak To Peak
Jitter Frequency range[11]	1 kHz to 110 MHz



Drive Jitter Injection

[11] *Jitter Injection Delay Modulation AWG sampling rate depends on active timing. Sample value range is -200ps..+200ps, some limitations do apply.*

PS9G Board ADC

The Board ADC provides very accurate high impedance voltage measurements behind each Pin Scale 9G channel. A Board ADC is shared among 16 channels.

Voltage range	-5.0 V to 8.0 V
Voltage resolution	300 uV
Accuracy	± 2 mV
Input impedance	> 10 MOhm

PS9G Driver

AC Performance

Transition time at 850 mV swing in 50 Ω (20 to 80%) [12]	65 ps ± 15 ps

DC Performance (specified into open and maximum output current ≤ 30mA)

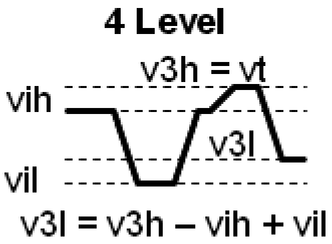
High level range	0.0 V to 2.5 V
Low level range	-0.8 V to 2.45 V
vt = v3h voltage level range	-0.5 V to 2.5 V
Maximum level shift (v3h voltage delta from Vih)	±850 mV
Swing range	50 mV to 1700 mV

DC Level accuracy @ 2 Level Mode

High voltage vih	±12 mV
Swing	±5 mV ± 1%
Differential Swing	±10 mV ± 1%

DC Level accuracy @ 3&4 Level Mode

High voltage vih	±20 mV if v3h ≤ 2V ±20 mV ± 0.1·(v3h - 2V) if v3h > 2V
Swing	±13 mV ± 1%
Differential Swing	±18 mV ± 1%
Termination voltage vt = Shifted High voltage v3h	±30 mV if v3h ≤ 2V ±30 mV ± 0.1·(v3h - 2V) if v3h > 2V
Shifted Swing	±15 mV ± 3%
Shifted Differential Swing	±20 mV ± 3%



DC Level accuracy @ 4 Level Mode

[12] measured for drive, no termination (v3h) switch

PS9G Receiver

AC Performance

intrinsic transition time (20 to 80%) at 850 mV swing in 50 Ohm	60 ps ± 20 ps

DC Performance Single Ended

Input Voltage and Threshold range	-0.5 V to 2.5 V
Threshold Voltage accuracy	±7 mV
Threshold Voltage resolution	±1 mV

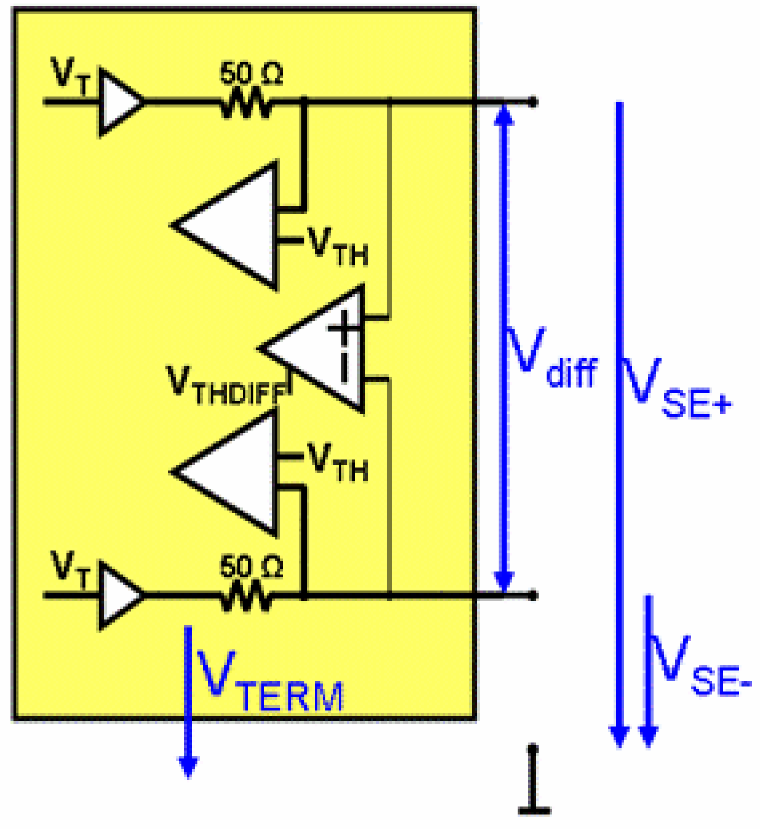
DC Performance Differential

Input Voltage range	-0.5 V to 2.0 V
Threshold Voltage range[13]	-850 mV to +850 mV
Threshold Voltage accuracy for 0 mV	±2 mV
Threshold Voltage accuracy[14]	±5 mV ±0.5% of Threshold
Threshold Voltage resolution	±1 mV

Termination

Single Ended / Center Tap Differential impedance	50 Ohm ± 2.5 Ohm
Termination level range	-0.1 V to 2.5 V
Termination Voltage accuracy	±7 mV
Termination Voltage resolution	±1 mV

[13] *Differential Threshold compares against the difference of VSE+ - VSE-*
[14] *with common mode voltage VCM =(VSE+ + VSE-)/2 in range 0..1V. If VCM outside 0..1V, Threshold Voltage accuracy ±7 mV ±1% of Threshold Voltage*


VT
50 Ω
VTH
VTHDIFF
+
-
VTH
VT
50 Ω
VTERM
Vdiff
VSE+
VSE-

PS9G Parametric Measurement Per-Pin PMU

PS9G Voltage force / measure

Voltage force / measure ranges

Range	-5.0 V to 8.0 V
Maximum input voltage range	-6.0 V to 9.0 V

Voltage force

Resolution	250 µV
Accuracy in range -5.0 V to 8.0 V	±(5 mV + I*R) [15]
Accuracy in range -0.0 V to 5.0 V	±(2 mV + I*R)

Compare mode

Resolution	500 µV
Accuracy in range -5.0 V to 8.0 V	±(2.5 mV + I*R) ± 0.5% of reading

Value measurement mode

Resolution	250 µV
Relative accuracy in range 0.0 V to 5.0 V[16]	±3 mV
Accuracy in range 0.0 V to 5.0 V	±(2 mV + I*R)
Accuracy in range -5.0 V to 8.0 V	±(5 mV + I*R)

Voltage Clamp

Accuracy	±(50 mV + I*R)

PS9G Current force / measure

	Range	Resolution	Measure accuracy	Force accuracy
Range 1[17]	±60 mA	5 µA	±90 µA ±0.2% of reading	±90 µA ±0.15% of setting
	±20 mA	5 µA	±90 µA ±0.1% of reading	±90 µA ±0.1% of setting
Range 2	±2 mA	0.5 µA	±2.0 µA ±0.1% of reading	±2.0 µA ±0.1% of setting

[15] *I is the actual current, R is the wiring resistance of ≤ 0.5 Ohm*

[16] *In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta voltage*

[17] *additional constraints apply for the sum of PPMU current of all channel; per board static(>1s):1A (=15.6mA per channel) dynamic (<1s):3A (=46.8mA per channel); per CC static(>1s):8A (=15.6mA per channel) dynamic (<1s):12A (=23.4mA per channel)*

	Range	Resolution	Measure accuracy	Force accuracy
Range 3	±200 µA	5 nA	±280 nA ±0.1% of reading	±300 nA ±0.1% of setting
Range 4	±20 µA	1 nA	±60 nA ±0.1% of reading	±60 nA ±0.15% of setting
Range 5	±5 µA	250 pA	±30 nA ±0.2% of reading	±30 nA ±0.2% of setting

Current force for relative measurement

Relative accuracy[18]	±30 µA ±0.3% of setting in range 1

Current Clamp

Accuracy	0 .. 15% of range

[18] *In two consecutive voltage value measurements with different force currents, the accuracy of the calculated delta of the forced currents*

PS9G Time Measurement Unit Per-Pin TMU

DC performance

Specifications of receiver apply (see *PS9G Receiver* on page 17).

AC performance

Maximum input frequency	4 GHz
Linearity error (see *PS9G Linearity* on page 12)	±5 ps
Accuracy, single timestamp to ARM (t0)	±35 ps
Accuracy, skew/propagation delay (between different pin resources)	2x (t_{EPA}+35ps)
Intrinsic jitter (RMS)	5 ps
Resolution	1 ps

Other (Maximum number of simultaneously active TMUs: 128 shared within slots 1-4 and slots 5-8)

Maximum Sample rate[19]	50 Mbps
Measurement Mode	Single ended & Differential

[19] *The maximum number of samples is limited by the available channel memory.*

PS9G Simultaneous Bi-Directional SBD

PS9G SBD OTA/EPA

The tester drive swing is one third the swing of device output, speed up to 5Gbps.

Typical Per-Pin Pair OTA/EPA

Drive Receive Pin Pair Overall Timing Accuracy (t_{OTA})	±40 ps
Drive Receive Pin Pair Edge Placement Accuracy (t_{EPA}=t_{OTA}/2)	±20 ps

Typical Per Board OTA/EPA

Overall Timing Accuracy (t_{OTA})	±70 ps
Edge Placement Accuracy (t_{EPA}=t_{OTA}/2)	±35 ps

PS9G SBD Eye Height

Single Ended (with Vswing = 500 mV into 50 Ohm, tester drive swing is one third the swing of device output)

	2.5 Gbps	4 Gbps	5 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	80 % of V_{swing}	65 % of V_{swing}	45 % of V_{swing}

Differential (with Vswing = 500 mV into 50 Ohm, tester drive swing is one third the swing of device output)

	2.5 Gbps	4 Gbps	5 Gbps
Typical $\Delta V_{SignalEyeHeight}$ of driver	85 % of V_{swing}	70 % of V_{swing}	70 % of V_{swing}
Typical $\Delta V_{SignalEyeHeight}$ of driver+receiver pair	83 % of V_{swing}	68 % of V_{swing}	50 % of V_{swing}

PS9G SBD Driver

AC Performance

Transition time at 500 mV swing in 50 Ohm (20 to 80%)[20]	85 ps ± 20 ps

DC Performance (specified into open and maximum output current ≤ 30mA)

High level range	0.0 V to 1.8 V with termination > 0.2 V 0.0 V to 1.5 V with termination ≤ 0.2 V

[20] *measured for drive, no termination (v3h) switch*

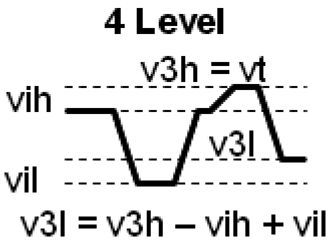
Low level range	-0.2 V to 1.45 V with termination > 0.2 V 0.0 V to 1.45 V with termination ≤ 0.2 V
vt = v3h voltage level range	0.0 V to 1.8 V
Maximum level shift (v3h voltage delta from Vih)	±600 mV
Swing range	50 mV to 1200 mV

DC Level accuracy @ 2 Level Mode

High voltage vih	±18 mV
Swing	±16 mV ± 5%
Differential Swing	±17 mV ± 4%

DC Level accuracy @ 3&4 Level Mode

High voltage vih	±25 mV if v3h ≤ 1.7 V ±25 mV ± 0.1·(v3h - 1.7 V) if v3h > 1.7 V
Swing	±15 mV ± 1%
Differential Swing	±20 mV ± 1%
Termination voltage vt = Shifted High voltage v3h	±30 mV if v3h ≤ 1.7 V ±30 mV ± 0.1·(v3h - 1.7 V) if v3h > 1.7 V
Shifted Swing	±20 mV ± 3%
Shifted Differential Swing	±25 mV ± 3%



PS9G SBD Receiver

AC Performance

intrinsic transition time (20 to 80%) at 850 mV swing in 50 Ohm	80 ps ± 20 ps

DC Performance Single Ended

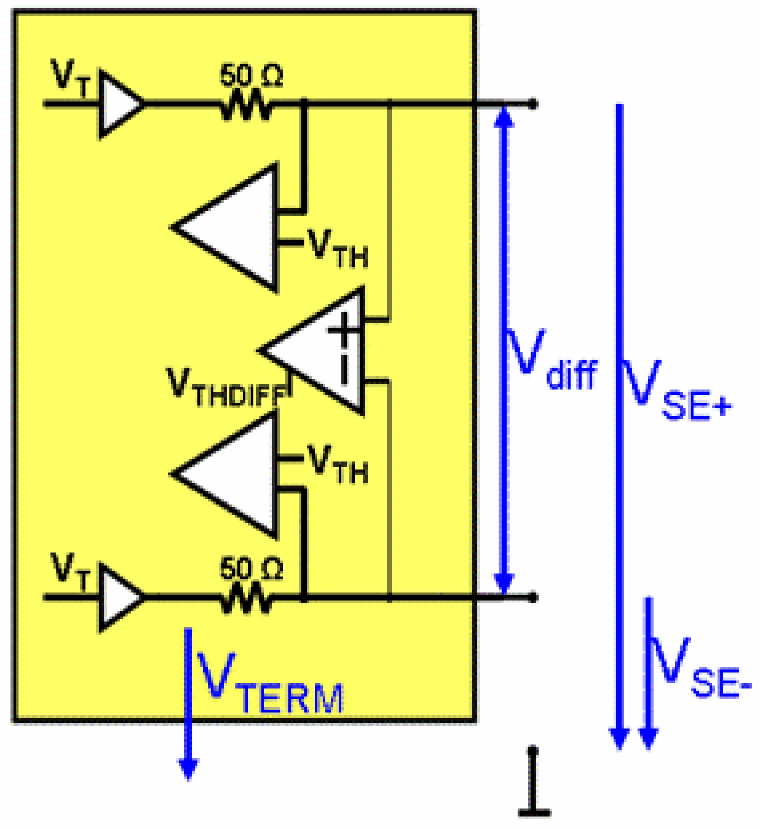
Input Voltage and Threshold range	0.0 V to 1.8 V
Threshold Voltage accuracy	±7 mV
Threshold Voltage resolution	±1 mV

DC Performance Differential

Input Voltage range	0.0 V to 1.8 V
Threshold Voltage range[21]	-850 mV to +850 mV
Threshold Voltage accuracy for 0 mV	±2 mV
Threshold Voltage accuracy[22]	±5 mV ±0.5% of Threshold
Threshold Voltage resolution	±1 mV

Termination

Single Ended / Center Tap Differential impedance	50 Ohm ± 2.5 Ohm
Termination level range	-0.1 V to 2.5 V
Termination Voltage accuracy	±7 mV
Termination Voltage resolution	±1 mV



[21] *Differential Threshold compares against the difference of VSE+ - VSE-*
[22] *with common mode voltage VCM =(VSE+ + VSE-)/2 in range 0..1V. If VCM outside 0..1V, Threshold Voltage accuracy ±7 mV ±1% of Threshold Voltage*